JETIR.ORG ISSN: 2349-5162 | ESTD Year : 2014 | Monthly Issue
JOURNAL OF EMERGING TECHNOLOGIES AND INNOVATIVE RESEARCH (JETIR)
An International Scholarly Open Access, Peer-reviewed, Refereed Journal

A Review on Polyherbal Handwash

Shaikh Nagma[1],Syed Rafe Noman[1],Shejule Shivbala[1], Shende Dhiraj[1], Kadam Chetan[2]*

1 Saraswati Institute of Pharmacy, Kuratdi Dist. Hingoli, Maharashtra, India

2 Saraswati Institute of Pharmacy, Kuratdi Dist. Hingoli, Maharashtra, India

(Pharmaceutics Department)

1 Students & 2Assistant Professor

Abstract:

The use of alcohol-based disinfectants for antibacterial hand washing has its fair share of drawbacks. These include unpleasant outcomes such as itching, dryness, irritation, and even dermatitis that may stem from synthetic hand wash preparations. To prevent such undesirable effects and allergic reactions, consumers are now searching for natural ingredient cosmetics. However, further research is needed before designing a hand wash that not only cleanses but also provides additional skin benefits.

In light of this, a liquid-based herbal hand wash was recently developed using extracts derived from Moringa Oleifera Leaves. The formulation was then subjected to various tests to evaluate its effectiveness in cleaning action while maintaining stability and other relevant parameters. Parameters tested included physical and chemical properties like pH levels, coloration, scent, appearance, texture, spread ability, grittiness factor as well as skin irritancy level measurement techniques along with foam height measurements.

The results showed that the herbal hand wash performed within normal limits with minimal or no adverse side effects whatsoever. This new development offers hope for those seeking natural ways to clean their hands without fear of harmful side effects caused by synthetic ingredients commonly found in conventional hand wash products. As research continues in this area, it is likely that more innovative solutions will be discovered to meet consumer demand for safe and effective personal care products.

Keywords: Herbal Handwash, Herbal extract, Shobhanjana, Hygiene, Cleaning, Foam retention,

Introduction:

The skin is the body's most exposed part and requires protection from skin pathogens. Herbal medicine has been utilized in treating and caring for a variety of illness [1]. Although various chemical antiseptics, such as alcohol-based sanitizers and chlorhexidine products, are now readily available on the market to prevent contagious disease transmission in healthcare settings more effectively, they have some drawbacks or side effects. If used frequently, these soaps or solutions can irritate the skin and make pathogens resistant [2]. This herbal hand wash

contains numerous natural herbs that are highly effective against certain microorganisms. Not only is herbal hand soap beneficial for both economic and medical purposes but microbes enter our bodies through our hands which makes using hand soap to wash your hands an essential step in preventing bacteria from entering one's system; thus protecting one's hands is just as important as stopping bacteria from getting in [3]. The scent of this herbal hand wash not only keeps the skin feeling fresh but also provides gentle frothing activity without causing any aggravation while utilizing natural ingredients to effectively remove oil and dirt from the skin along with aiding in removing antiseptic and fungal skin issues. Therefore, regular hand washing aids in preventing diarrhea, respiratory infections, eye infections along with ensuring overall good health by avoiding transmitting pathogenic microorganisms that cause illness making it a simple yet[4] least expensive measure to prevent infection especially during current COVID-19 pandemic conditions where prevention is better than cure; Hand hygiene plays a vital role here since it involves cleaning hands with purposeful intent of removing soil, dirt, pathogenic microorganisms while keeping transient organisms away at bay[5]

Advantages and disadvantages of herbal handwash: -

Advantages [6]

1. Absence of adverse effects.
2. Reduction in the number of bacteria present on hands.
3. Provides antiseptic and fungal relief to skin problems.
4. Effectively removes dirt and oil from the skin.
5. More convenient than using soap and water.
6. The most effortless way to eliminate microorganisms.
7. Prevents germs from entering our body.

Disadvantages

1. Prolonged skin harm.
2. Irritative contact dermatitis and eczema.

Benefits of Herbal Handwash [7]

- Convenient availability
- Cost-effective
- Enhanced efficacy in promoting hand hygiene
- Reduced occurrence of side effects compared to other types of hand washes.

The Materials Utilized in the Formulation of Our Herbal Handwash:

1. Moringa oleifera
2. Carbopol 940
3. Methyl paraben

4. Glycerine
5. Tulsi
6. Pudina
7. Clove oil
8. Ritha
9. SLS
10. Jasmine oil
11. Distilled water.

These ingredients were carefully selected based on their Pharmacognosy properties derived from various herbal plants.

Pharmacognosy of Herbal Plants

1. Moringa oleifera:

Family: Moringaceae

Genus: Moringa

Common names: Moringa, Drumstick tree, Horseradish tree, and Benzo live tree.

Hindi name: Sainjna, Saguna

Marathi name: Shevga

Synonym: Morning peregrina.

Sanskrit name: Shobhanjana.

Moringa (moringaoleifera Lann).is type oflocal medicinal Indian herb which has turn out tobe familiar in the tropical and subtropical countries. Moringaoleifera Is one of the vegetables of theBrassica order and belong to the familyMoringaceae. It was recently discovered that the Moringa Oleifera seed extract exhibited antifibrotic effects on liver fibrosis in rats (Hamza ,2010).

2. Tulasi:

Family: Lamiaceae

Genus: Ocimum

Common name: Holy basil, Tulasi, and tamole

Hindi name: Holy basil, pushpsara, Nandini.

Marathi name: Tulasi.

Sanskrit name: Vrinda.

Synonyms: Gauri bahumanjari (L.) Merr.

Medicinal use of tulasi

To treat ring worm and other skin diseases like leucoderma, a paste made from tulsi leaves is applied to the affected area. Saffron and tulsi leaves are combined with chickenpox to investigate the condition. In cases of normal glucose, the ethanolic extract of tulsi leaves results in a significant drop in blood sugar.

3. Ritha:

Family: Sapindaceae

Genus: Sapindus

Common name: Soapberry, soapnut, washnut, aritha, dodan, and dodani.

Hindi name: soap nut, aritha

Marathi name: soap nut tree, ritha

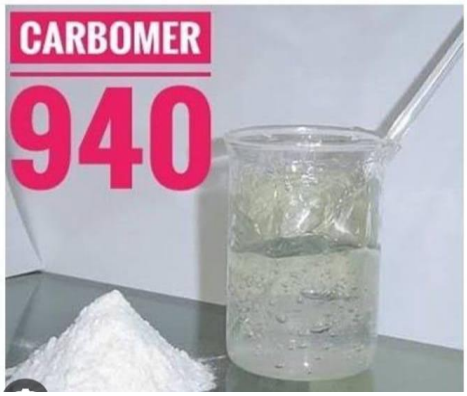
Sanskrit name: Arishtah.

Synonym: soapberry, soapnut, washnut, aritha, dodan, and Indian soap berry.

4. Carbopol 940

Carbopol 940 category:

Carbopol® 940 polymer is a white powder, crosslinked polyacrylic acid polymer. It is an extremely efficient rheology modifier capable of providing high viscosity and forms sparkling clear gels or hydro-alcoholic gels and creams.



Generic name:

Carbomer 940, also called Carbomer or Carboxypoly-methylene is a generic name for synthetic high molecular weight polymers of acrylic acid used as thickening, dispersing, suspending and emulsifying agents in pharmaceuticals and cosmetics.

Formulation Table:

Sr. No.	Ingredient	Quantity (gm/ml,)	Action
1	**Moringa oleifera**	**10**	**Antimicrobial agent**
2	**Tulsi**	**10**	**Purifying agent**
3	**Pudina**	**5**	**Antibacterial agent**
4	**Clove oil**	**0.50**	**Antibacterial agent**
5	**Ritha**	**5**	**Foaming agent**
6	**SLS**	**3**	**Foaming agent**
7	**Carbopol 940**	**5**	**Gelling agent**
8	**Methyl paraben**	**0.50**	**Preservative**
9	**Glycerine**	**0.5**	**Softening agent**
10	**Jasmine oil**	**Q.S.**	**Perfume**
11	**Distilled water**	**Upto 100 ml**	**Vehicle**

Method of Preparation [8][9]

1) Polyherbal Hand wash Gel was prepared using Carbopol940 as Gelling agent which is socked in 15ml distilled water overnight.

2) Neem and Peppermint extracts, Ritha Powder along with Tulsi and Clove oil were measured accurately and dissolved by gentle heating.

3) After heating, keep the solution aside for sometimes.

4) The required quantity of Sodium lauryl Sulphate dissolved in 10ml distilled water along with Glycerine were mixed in above aqueous phase with continuous stirring.

5) The methyl paraben was dissolved in remaining quantity of purified water and dispersed into the extract.

6) The swelled polymer (Carbopol 940) was stirred using a mechanical stirrer to ensure the uniform dispersion of polymer and finally added into the above mixture to form a Homogenous Gel and then the required quantity of Rose oil was added for Fragrance.

7) Lastly, it was stored in well closed container and labelled suitably for further analysis

Evaluation parameters:

1) **Organoleptic Evaluation Parameters:** like color, odour, texture was carried out Color and textures were evaluated by visual and touch sensation respectively. The odour was inspected by sensing the formulation. [8][9][10][11]

2) **Appearance and Homogeneity:** Appearance and Homogeneity was evaluated by visual inspection.[9][10][12]

3) **Grittiness:** 1ml of Gel was taken on finger tips and rubbed between two fingertips, then the formulation was evaluated.[10][12][13]

4) **Skin Irritation Test:** Skin Irritation Test was evaluated by applying Polyherbal Hand wash Gel on skin and left for 30 min, after 30 minutes of washing observe any itching, rashes, or redness on skin by sensory and visual inspection. [8][9]

5) **pH:** 1gm of Sample of Polyherbal Hand wash Gel was taken and dissolved it into 100ml distilled water. The pH solution was measured by standardized digital pH meter. [8][9][12]

6) **Spread ability**: 0.5gm of Sample of Polyherbal Hand wash Gel was pressed between two slides and left for about 5 minutes where no more spreading was expected. Diameter of spreaded circle was measured in cm and was taken as comparative values for spread ability [9]

7) **Viscosity**: The viscosity of Polyherbal Hand wash Gel was determined by using Ostwald viscometer.[9][12]

Conclusion: -

It was concluded that the herbal hand wash which are prepared from herbal ingredients they show fewer side effect as compared to synthetic hand wash which are prepared from synthetic compound.The prepared herbal hand wash was evaluated using various parameter and was found to be satisfied for the use.

Reference:

1) Mr. Bhise Akash Bhagwan,Formulation& Evaluation of Herbal Hand Wash By using Natural Ingredient By simple Method,IGCRT,9(12),2021 page no.b627
2) Sandeep DS,etal , Formulation of antimicrobial polyherabal hand wash RJPT,9(7)2016 ,page no.1
3) Ramanamma lalam ,etal ,Formuation&evaluation of herbal hand wash ISSN,11(12) page no.182
4) Aman Shukla ,etal , Formulation & evaluation of herbal hand wash using Ginger Rhizomes Ijppr.Human 2020 ,19(1) ,page no.124
5) Natarajan SB, Shah MA. Formulation Evaluation and Antibacterial Efficiency of Herbal Hand Wash Gel. International Journal of Pharmaceutical Science 23, 2014, 120-124.

6) Mr. Bhise Akash Bhagwan, Formulation & Evaluation of Herbal Hand Wash By using Natural Ingredient By simple Method, IGCRT,9(12),2021 page no. b629.

7) Bahuguna M, Kashyap S. Formulation and evaluation of hand wash. World Journal of PharmaceuticalResearch. 2016 May 19; 5(7):1559-77

8) Shaloo, Shayna Shafi, Simran Singh, Shashi Verma, Ritesh Kumar Tiwari and Devika Tripathi, Formulation and Development of Mint containing Herbal Hand Sanitizer, European Journal of Pharmaceutical and Medical Research, 2017 [11], 454 - 457.

9) Powar P.V, Bhandari N. R, Arya Ashwini, Sharma P. H., Formulation and Evaluation of Poly Herbal Anti - Bacterial Gel Based Hand Wash, International Journal of Pharmaceutical Sciences Review and Research, 33 (1) July - August 2015; Article No.16, Pages: 79 - 82.

10) Mounika, Vijayanand P, V. Jyoti, Formulation and Evaluation of Polyherbal Hand wash Gel containing essential oils, International Journal of Pharmacy and Analytical Research, Vol.6, Issue 4, Oct - Dec 2017, P. N.645 - 653.

11) Zeeshan Afsar, Salma Khanam, Formulation and Evaluation of Polyherbal Soap and Hand Sanitizer, International Research Journal of Pharmacy, 2016, 7 [8]

12) Andeep D.S, NarayanaCharyulu R, Prashant Nayak, Aliss Maharjan, Indira Ghalan, Formulation of Antimicrobial Polyherbal Hand Wash, Research J. Pharm and Tech., July 2016, 864 - 866 [13] Megha Bahuguna and Shil1pi Kashyap, Formulation and Evaluation of Hand Wash, World Journal of Pharmaceutical Research. Volume 5, Issue 7, 1559 - 1577